Aryan Harkawat

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Education

University of Texas at Austin — Austin, TX
Bachelor of Science in Computer Science — *May 2028*

- **Coursework:** Operating Systems, Computer Architecture, High Performance Computing, Algorithms, Data Structures, Linear Algebra, Statistics, Machine Learning

Work Experience

Stripe — Jan 2027 – May 2027
Incoming Software Engineer Intern

Stripe — May 2026 – Aug 2026
Software Engineer Intern, Core Infrastructure

- Rebuilt how Stripe's service mesh picks a backend so traffic avoids overloaded servers (power-of-two-choices on reported CPU and queue depth): overload errors down 40%, ∼$200K/yr saved, 400M+ requests/day
- Added zone-aware routing that spills traffic to a neighboring zone once the local one passes 80% utilization, cutting p99 latency 23% (300ms → 230ms), rolled out behind feature flags with zero incidents

Trend Micro — Aug 2025 – May 2026
Software Automation Intern

- Built a scheduled Python ETL that reconciles payroll records across Excel and shared drives and flags exceptions
- Built a Python/Power Automate pipeline that sessionizes raw network logs into SQL and flags anomalies

Mercor — Feb 2025 – Apr 2025
Machine Learning Intern

- Wrote the evaluation harness (Hugging Face Transformers) that scored 6 fine-tuned BERT ticket classifiers on precision, recall and F1, so picking a model took an hour instead of a day of manual runs
- Exported the chosen model to ONNX with int8 quantization for 3× faster CPU inference (∼140 tickets/s)
- Wrote Pandas checks that caught duplicate, missing and conflicting labels in 1M+ tickets, raising F1 by 2 points

Personal Projects

Concertify | *Go, gRPC, Python, PyTorch, Node.js, Redis, Prometheus, Docker* | Demo · Source

- Concert recommender with 500+ users: reads your Spotify listening history, finds nearby Ticketmaster shows and ranks them by artist overlap and genre similarity, with a diversity re-rank so the list isn't all one genre
- Ranking runs in a Go gRPC service (∼1 ms/request), per-user weights are fit in PyTorch from thumbs up/down
- Cut cold start 9s to 2.5s under Ticketmaster's 5 req/s limit with a deduping Redis cache and a priority rate limiter

Herd | *Go, Envoy, gRPC, Kubernetes, Prometheus, Docker* | Source

- Gateway in front of self-hosted LLM servers (OpenAI-compatible API) that sends each request to the replica with the most free KV-cache and a warm prefix cache, cutting p99 time-to-first-token 55% vs. round-robin
- Sheds low-priority requests under load, queues fairly across tenants and spills to a cloud provider within a budget: with two degraded replicas p99 TTFT stayed at 12.6s vs. 292s for round-robin
- Runs as an Envoy ext_proc endpoint picker, with Kubernetes manifests, a load generator and a live dashboard

Volumetric Cloud Renderer | *C++, OpenGL, GLSL, CMake* | Source

- Real-time clouds in C++/OpenGL: ray-marches 3D Perlin-Worley noise generated on the GPU through a curved cloud layer, lit with a multiple-scattering approximation
- Half-res cloud pass plus reprojection of the previous frame (3.5× cheaper), headless renders and CI benchmarks

Technical Skills

Languages: Go, Python, C++, C, Java, SQL, TypeScript, JavaScript, Bash, GLSL
Infrastructure: gRPC, Protobuf, Envoy, Kubernetes, Docker, Redis, Prometheus, Linux, GitHub Actions, AWS
ML & Other: PyTorch, Hugging Face Transformers, ONNX, Pandas, NumPy, OpenGL, CMake, React, Node.js